

MENDE, France

WMO: 075540

Lat: 44.5336N

Lon: 3.4544E

Elev: 2349

StdP: 13.49

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.1	(c) 19.1	(d) 5.0	(e) 7.7	(f) 22.2	(g) 10.1	(h) 10.1	(i) 25.7	(j) 24.8	(k) 29.8	(l) 21.0	(m) 33.4	(n) 6.8	(o) 300	(p) 0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	86.7	63.7	83.3	62.8	80.1	61.7	66.3	81.2	64.7	78.5	63.2	76.3	8.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	88.1	71.3	59.7	83.3	70.1	58.2	79.0	68.4	32.4	81.3	31.0	78.1	29.9	76.3	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min		
(a) 19.1	(b) 16.6	(c) 14.5	(d) 11.0	(e) 91.4	(f) 4.9	(g) 4.0	(h) 7.5	(i) 94.2	(j) 4.6	(k) 96.5	(l) 1.8	(m) 98.8	(n) -1.7	(o) 101.6		
(a) 19.1	(b) 16.6	(c) 14.5	(d) 11.3	(e) 68.9	(f) 5.6	(g) 1.3	(h) 7.2	(i) 69.9	(j) 3.9	(k) 70.6	(l) 0.8	(m) 71.3	(n) -3.3	(o) 72.3	(p) 72.3	(q) 72.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	34.5	35.2	42.1	47.2	52.5	60.9	65.8	65.3	58.7	51.6	43.0	37.8
	DBStd	12.70	6.35	8.32	6.34	6.02	6.64	6.87	5.88	6.16	5.86	6.85	6.97	6.94
	HDD50	1994	480	415	252	125	49	2	0	0	5	66	221	379
	HDD65	5827	945	834	710	534	387	159	62	76	201	416	659	844
	CDD50	1863	0	1	7	41	128	330	490	473	265	116	12	0
	CDD65	222	0	0	0	0	1	38	88	84	11	0	0	0
	CDH74	2559	0	0	0	4	34	440	960	938	170	13	0	0
	CDH80	709	0	0	0	0	2	115	277	293	22	0	0	0
(14) Wind	WSAvg	6.4	6.1	6.7	7.3	7.0	7.1	6.3	6.3	5.9	5.7	6.2	6.0	6.0
(15) Precipitation	PrecAvg	34.9	2.3	2.0	1.8	3.3	3.5	2.9	2.4	2.8	3.6	3.8	3.9	2.6
	PrecMax	45.6	6.0	5.0	6.5	8.1	6.9	9.1	5.7	8.5	9.9	7.1	7.6	7.0
	PrecMin	23.6	0.0	0.2	0.0	0.2	0.9	0.4	0.8	0.6	0.7	1.5	1.3	0.2
	PrecStd	5.6	1.4	1.2	1.5	1.9	1.7	1.5	1.1	1.5	2.0	1.5	1.8	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	63.0	67.5	72.1	78.2	89.1	90.5	90.4	83.0	75.6	65.1	56.5
		MCWB	43.9	46.9	47.9	54.2	58.3	66.4	66.0	63.5	59.9	55.8	53.4	44.7
	2%	DB	48.9	56.4	62.5	67.5	73.5	82.8	86.2	87.0	78.5	70.1	58.2	51.7
		MCWB	43.7	43.4	46.6	52.2	57.4	63.6	63.9	63.1	58.8	55.6	50.0	43.1
	5%	DB	46.1	49.8	58.0	63.7	69.8	78.9	82.8	83.4	75.2	65.7	54.5	48.7
		MCWB	42.2	41.6	44.7	50.2	56.3	62.0	62.9	62.8	58.1	54.4	49.6	44.5
	10%	DB	43.8	45.0	52.8	59.6	65.5	75.2	79.7	79.6	71.4	61.7	52.2	46.7
		MCWB	40.8	40.2	43.3	48.4	53.9	60.3	62.0	61.7	57.1	54.0	48.6	43.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	48.2	49.8	56.0	61.3	68.5	68.2	67.9	63.5	60.2	54.4	48.6
		MCDB	51.3	60.2	61.0	69.5	73.8	83.8	85.6	83.8	75.1	67.5	62.1	51.3
	2%	WB	45.0	44.8	48.0	53.2	58.8	65.4	66.4	65.9	62.0	58.0	52.0	46.6
		MCDB	47.8	51.5	58.9	65.5	70.8	79.6	81.4	80.2	73.3	63.9	56.5	48.7
	5%	WB	43.0	42.7	46.2	51.2	56.6	63.1	64.7	64.4	60.3	56.4	50.4	45.4
		MCDB	45.6	48.0	54.0	61.6	67.6	76.0	78.6	78.2	70.7	62.0	53.6	47.7
	10%	WB	41.2	40.8	44.5	49.0	54.7	61.2	63.2	62.9	58.5	55.1	49.0	43.7
		MCDB	43.3	44.7	51.3	57.5	64.1	73.4	76.5	75.8	68.1	60.9	51.9	46.3
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	14.2	17.0	17.7	18.3	20.6	22.3	23.0	20.4	16.7	11.8	12.2
		MCDBR	16.3	23.7	26.4	26.0	25.0	26.5	28.1	29.1	27.5	24.5	16.5	16.2
		MCWBR	11.4	14.0	13.8	13.6	12.7	12.0	11.3	10.9	12.0	12.7	10.6	11.0
	5% WB	MCDBR	13.4	19.0	21.0	23.2	22.6	24.9	24.5	24.3	22.2	17.4	13.4	11.5
		MCWBR	10.4	12.3	12.1	12.5	12.3	12.5	11.2	10.9	10.8	10.2	9.4	8.8
(40) Clear-Sky Solar Irradiance	taub		0.283	0.302	0.338	0.372	0.380	0.382	0.376	0.374	0.350	0.341	0.308	0.282
	taud		2.507	2.464	2.394	2.304	2.328	2.351	2.364	2.381	2.429	2.462	2.507	2.521
	Ebn at Noon		270	283	285	282	282	281	281	279	277	264	257	259
	Edh at Noon		21	27	33	39	39	39	37	36	31	27	21	19
(44) All-Sky Solar Radiation	RadAvg		464	749	1111	1428	1713	2033	2030	1767	1356	841	505	412
	RadStd		56	88	127	173	171	154	158	116	96	93	65	55

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page